

Sigma 6.0 (DSA)

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%HASHING SOLUTIONS

Solution 1:

Using the concept of Horizontal Distance (HD), as discussed in Top View of a Binary Tree Question in the Trees chapter.

```
import java.util.*;

class Solution {
    static class Node {
        int data;
        int hd;
        Node left, right;

        public Node(int key) {
            this.data = key;
            this.hd = Integer.MAX_VALUE;
            this.left = this.right = null;
        }
    }

    public static void bottomViewHelper(Node root, int curr, int hd,
    TreeMap<Integer, int[]> m) {
        if (root == null)
            return;

        // If node for a particular HD is not present, add to the map.
        if (!m.containsKey(hd)) {
            m.put(hd, new int[]{ root.data, curr });
        }

        // Compare height for already
        // present node at similar horizontal
        // distance
        else {
            int[] p = m.get(hd);
            if (p[1] <= curr) {
```